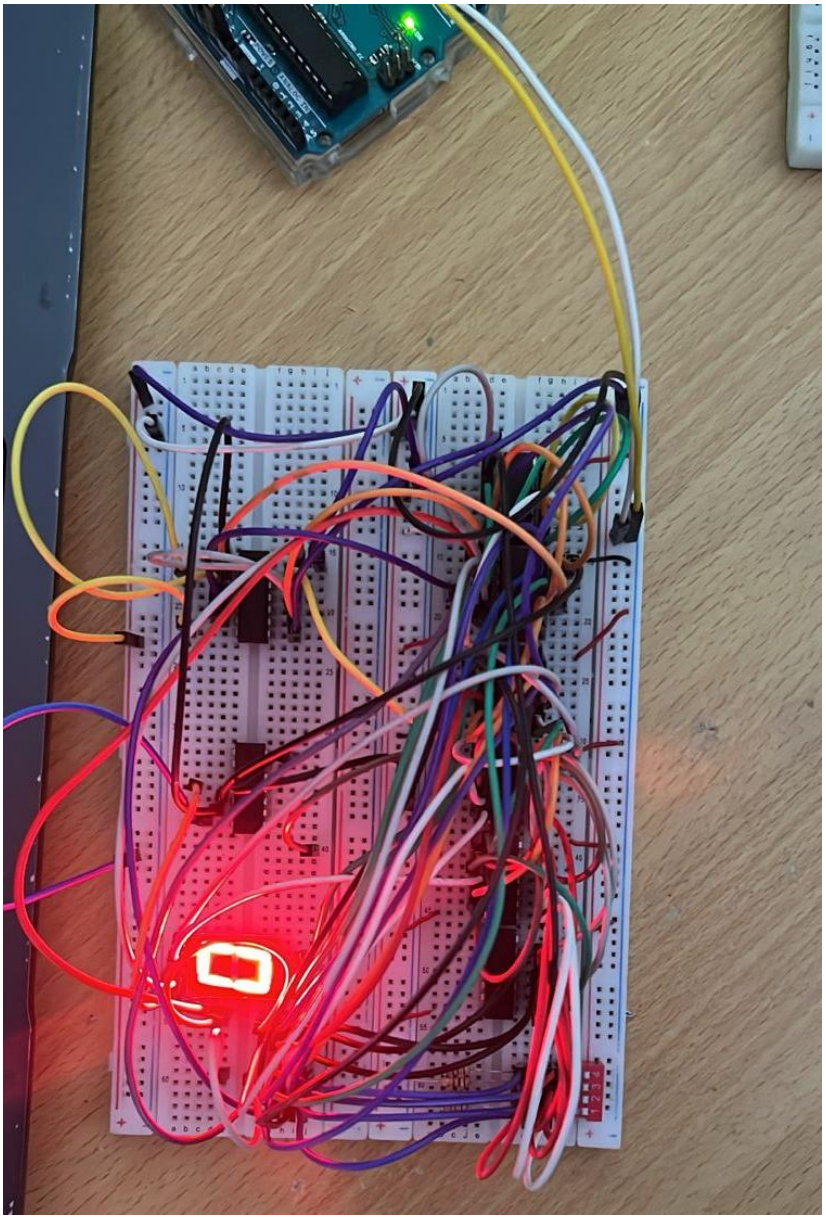


A CIRCUIT FOR BCD (8-4-2-1) TO 7-SEGMENT DISPLAY

ABSTRACT

This project demonstrates the design and implementation of a BCD to 7-segment display circuit without using a dedicated decoder IC. Instead, basic logic gates are used to realize the combinational logic required to convert a 4-bit BCD input (0000-1001) into signals that drive the seven segments (a-g) of a display to represent digits 0-9

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DIPLOMA IN ENG TECH: ELECTRICAL ENGINEER

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Introduction

In digital electronics, numbers are often represented in Binary Coded Decimal (BCD) form. A 7-segment display is convenient device for showing these numbers visually .Normally, a BCD -to-7-segment decoder IC (like 7447) is used ,but in this project we manually designed and implemented the logic using basic logic gates. This helps in understanding how combinational circuits works and how binary inputs are converted into human-readable numeric displays.

Objective

To design and implement a BCD to 7-segment display circuit using only basic logic gates (AND, OR, NOT, NAND, NOR) without using a dedicated decoder IC.

Theory

The display has 7 LEDs labelled **a-g,** arranged to form numbers. Each segment is turned ON and OFF to show digits 0-9. The truth table for each segment was used and the Karnaugh maps to help minimize the Boolean expression by grouping adjacent "1" cells and deriving the simplest sum-of-products expressions.

TABLE TRUTH

A	B	C	D	a	b	c	d	e	f	g
0	0	0	0	1	1	1	1	1	1	0
0	0	0	1	0	1	1	0	0	0	0
0	0	1	0	1	1	0	1	1	0	1
0	0	1	1	1	1	1	1	0	0	1
0	1	0	0	0	1	1	0	0	1	1
0	1	0	1	1	0	1	1	0	1	1
0	1	1	0	1	0	1	1	1	1	1
0	1	1	1	1	1	1	0	0	0	0
1	0	0	0	1	1	1	1	1	1	1
1	0	0	1	1	1	1	1	0	1	1

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Appropriate Karnaugh maps

AB\CD	00	01	11	10
00	1	0	1	1
01	0	1	1	1
11	0	0	0	0
10	1	1	0	0

$a = A + C + B.D + \bar{B}\bar{D}$

AB\CD	00	01	11	10
00	1	1	1	1
01	1	0	1	0
11	0	0	0	0
10	1	1	0	0

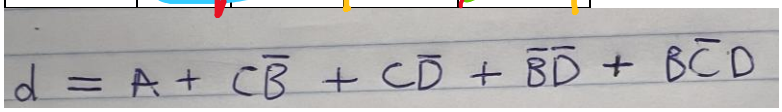
$b = \bar{B} + CD + \bar{C}\bar{D}$

AB\CD	00	01	11	10
00	1	1	1	0
01	1	1	1	1
11	0	0	0	0
10	1	1	0	0

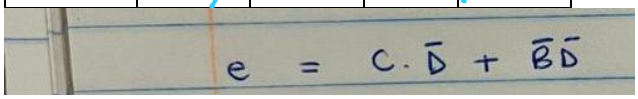
$c = B + D + \bar{C}$

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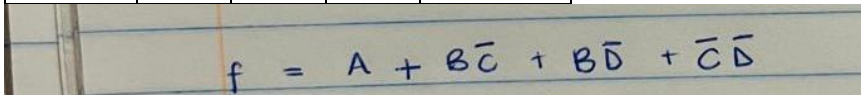
AB\CD	00	01	11	10
00	1	0	1	1
01	0	1	0	1
11	0	0	0	0
10	1	1	0	0



AB\CD	00	01	11	10
00	1	0	0	1
01	0	0	0	1
11	0	0	0	0
10	1	0	0	0

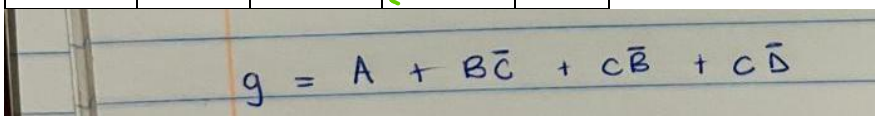


AB\CD	00	01	11	10
00	1	0	0	0
01	1	1	0	1
11	0	0	0	0
10	1	1	0	0



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AB\CD	00	01	11	10
00	0	0	1	1
01	1	1	0	1
11	0	0	0	0
10	1	1	0	0



- **Components Used**

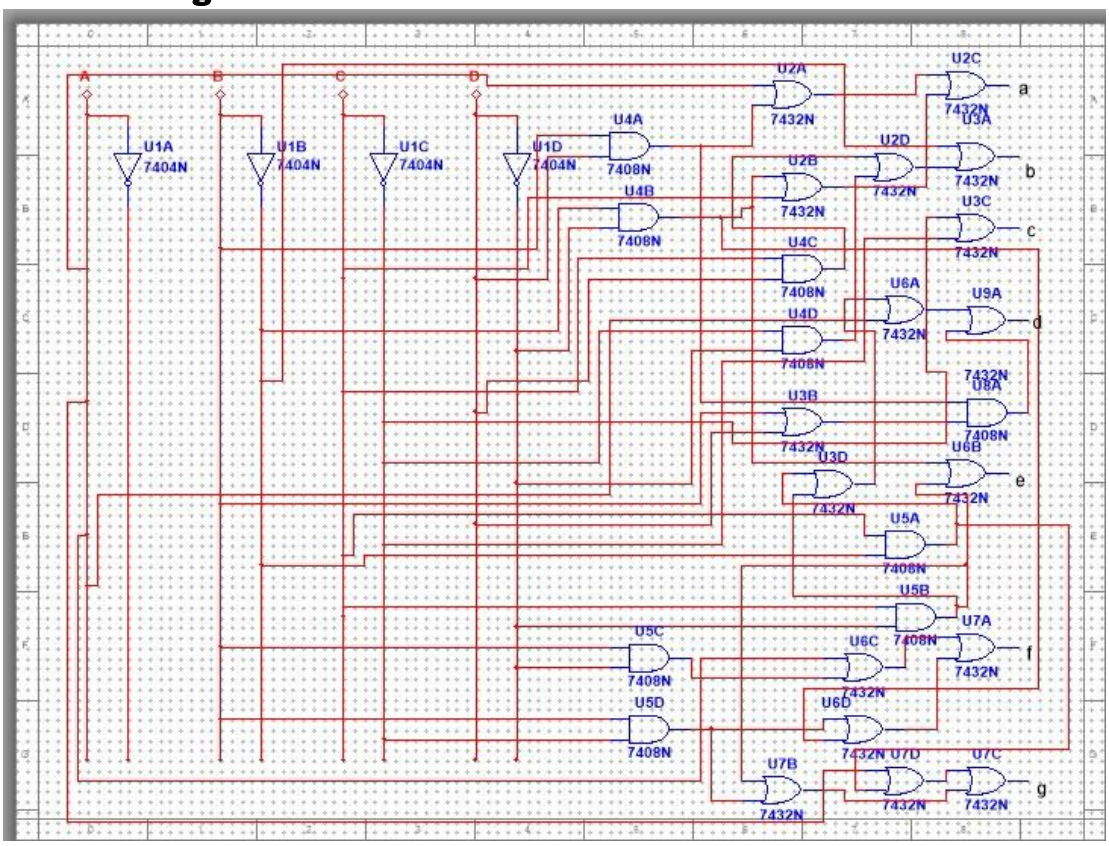
Components	Quantity	Description
Logic Gate ICs	As required	For combinational logic
7-Segment Display	1	
Resistors (1K)	4	Current limiting for LED
Breadboard	3	For circuit assembly
Wires	_	Connections
Power Supply	1	5 V DC

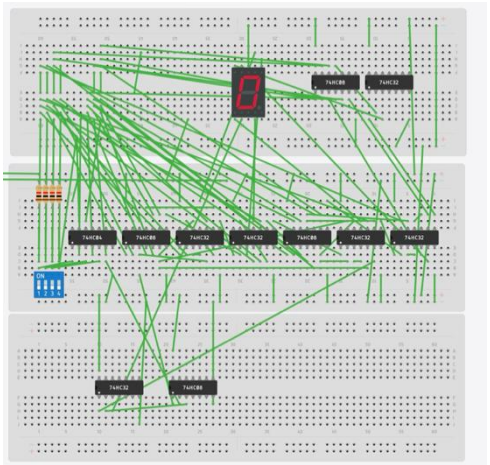
Simulation Overview & Working

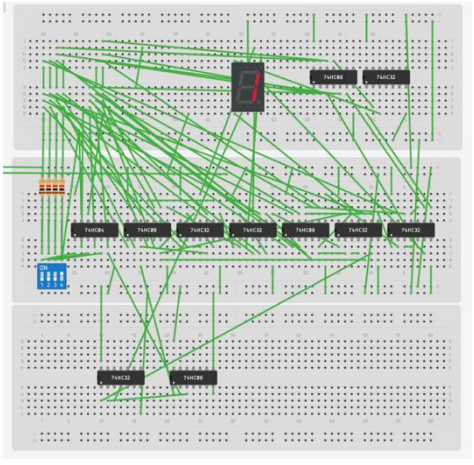
The circuit composed from the simplified Karnaugh maps for each segment was build using Multisim. To verify the correct operation of the BCD to 7 segment display before the hardware implantation, the circuit was simulated using Tinkercad. The BCD input is applied to the four inputs lines (A, B, C, D). The logic gate network for each segment generates HIGH and LOW outputs based on these inputs, lighting the correct segments to form the corresponding digits on the display. Each digit from 0-9 lights up correctly according to the truth table. The simulation results matched the theoretical truth table.

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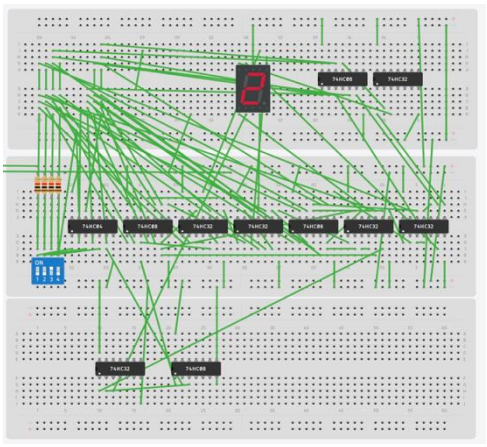
Circuit Design and Simulation

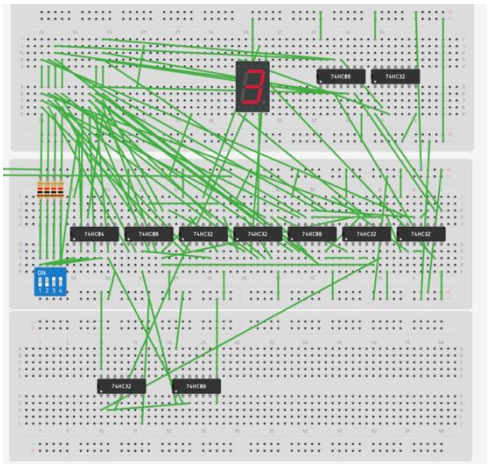


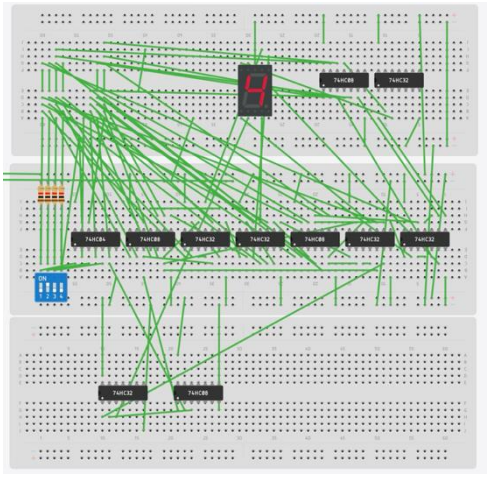


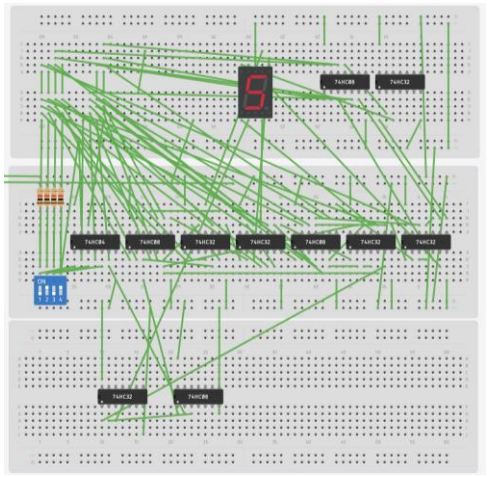


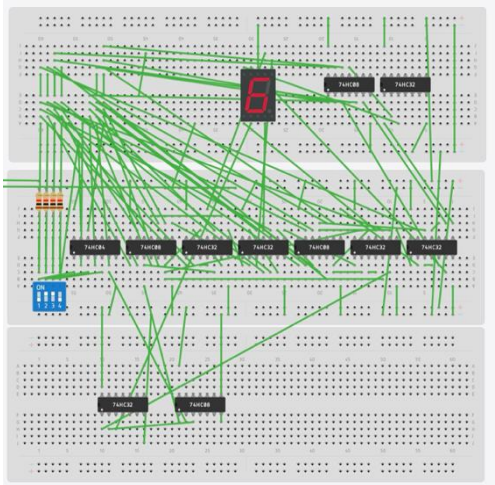
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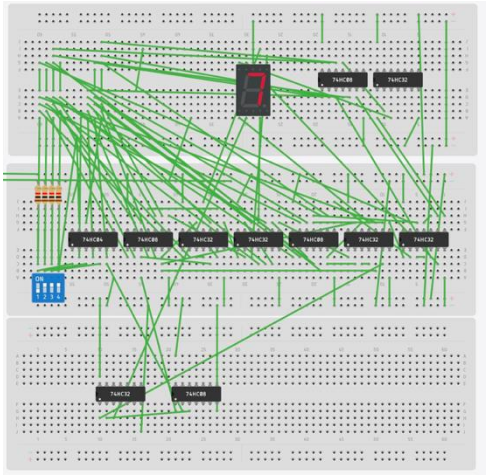




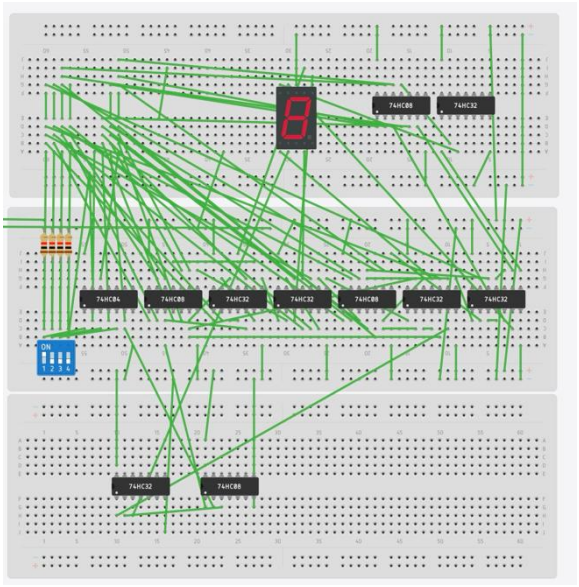


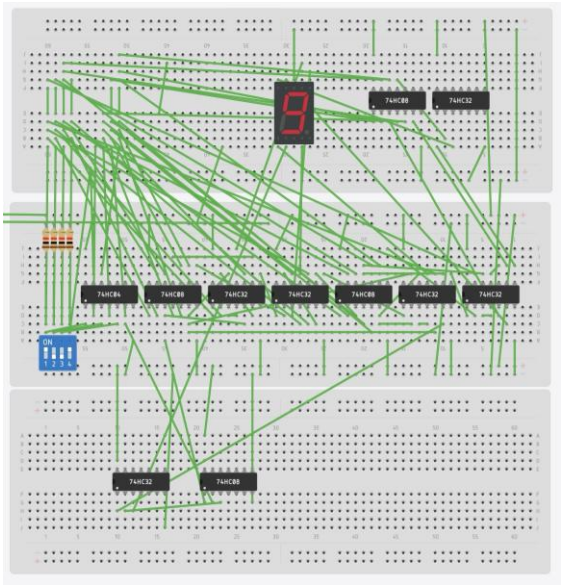






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Applications

- Digital counters
- Clocks and timers

Conclusion

The project successfully demonstrate how a BCD to 7-segment display can be implemented using only basic logic gates. This manual implementation helps understand combinational circuit design, Boolean simplification, and practical display interfacing without using a decoder IC. We gained a deeper understanding of how the display operates at a low level. This approach reinforced our knowledge of binary numbers representation, segment mapping and control logic.

References

- *Floyd, T.L., Digital Fundamentals, 11th Ed, Pearson Education Limited 2015*